

Worksheet · Section 3.11

Binary Search



The Big Picture

Binary search finds values fast by cutting the list in half each step — but only on sorted data. Practice tracing it, counting checks, and choosing the right search.

Part 1 · Vocab Match

Write the letter of the matching definition on each line.

- | | |
|-------------------------------|--|
| _____ 1. Binary search | A. One check of an element against the target |
| _____ 2. Linear search | B. Arranged in order |
| _____ 3. Sorted | C. The middle element checked at each step of binary search |
| _____ 4. Midpoint | D. How much time or resources an algorithm uses as the input grows |
| _____ 5. Worst case | E. Checks elements one at a time from the start |
| _____ 6. Efficiency | F. Remove from consideration, like half of the list |
| _____ 7. Eliminate | G. The input that takes the most steps |
| _____ 8. Comparison | H. Checks the middle of a sorted list and eliminates half each step |

Part 2 · Fill It In

Use each word from the word bank once.

middle

comparisons

1

linear

half

10

sorted

unsorted

1. Binary search only works on _____ data.
2. Each step of binary search eliminates about _____ of the data.
3. Binary search starts by checking the _____ element.
4. _____ search checks elements one at a time from the start.
5. Doubling a sorted list adds about _____ step to binary search.
6. Binary search needs at most about _____ checks for 1,000 items.
7. Linear search works on _____ lists.
8. Search efficiency is measured by counting _____.

Name: _____

Class: _____

Date: _____

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Practice

Part 3 · Trace a Binary Search

List (10 elements, positions 1–10). When the middle falls between two elements, use the left one.

[2, 5, 8, 12, 16, 23, 38, 56, 72, 91]

Target	Values checked, in order	Found?
23	_____	_____
91	_____	_____
4	_____	_____

Part 4 · Count the Checks

Sorted list size	Binary (worst case)	Linear (worst case)
16	_____	_____
100	_____	_____
1,000	_____	_____
1,000,000	_____	_____

Pattern: when n doubles, binary search needs _____ more check.

Part 5 · Which Search?

Situation	Best search
Finding a name once in an unsorted sign-up sheet	_____
Looking up a word in a sorted dictionary of 500,000 words	_____
Checking a list of only 5 items	_____
A sorted list of 1 million IDs searched thousands of times a day	_____

Part 6 · Explain It

a) Why must data be sorted for binary search to work?

b) When could linear search finish FASTER than binary search?



Think:

Binary search halves the list each step: $1,000 \rightarrow 500 \rightarrow 250 \rightarrow 125 \rightarrow 63 \rightarrow 32 \rightarrow 16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1$. That's about 10 checks instead of 1,000!

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Exam Practice

Part 7 · Exam Practice

★ Warm-up

1. What must be true before using binary search?
- A. The list must have an even length
 - B. The list must be short
 - C. The list must be sorted
 - D. The list must contain only numbers

★★ On Level

2. What is the maximum number of checks binary search needs for a sorted list of 64 items?
- A. 32
 - B. 64
 - C. 7
 - D. 6

★★ On Level

3. Using binary search on [1, 3, 5, 7, 9], which element is checked first?
- A. 3
 - B. 5
 - C. 1
 - D. 9

★★★ Exam Hard

4. A sorted list grows from 500 items to 4,000 items. About how many more checks does binary search need in the worst case?
- A. It doubles
 - B. 3,500
 - C. 3
 - D. 8

★★★ Exam Hard

5. Which TWO statements are true? (*Select two.*)
- A. Linear search requires sorted data.
 - B. Binary search works on any list.
 - C. For large sorted lists, binary search usually needs far fewer checks.
 - D. Binary search can be slower than linear search when the target is the first element.

Part 8 · Written Response

A teacher stores 2,000 student IDs. Compare linear and binary search for finding one ID: give the worst-case checks for each and explain what must happen first for binary search.
